

AMLTa - COMPARATIVE ANALYSIS

AI-Based Early Detection of Alzheimer's Disease Using Speech and Machine Learning

Section: 1

Team : 3

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Comparative Analysis

The proposed model (Experiment 8: DeBERTa-v3-base with attention pooling and mean pooling concatenated before a dropout of 0.30 and a binary classifier) is compared with published transcript-based systems on five metrics: accuracy, precision, recall (sensitivity), F1 score and specificity. On the held-out 75-transcript test set (37 healthy, 38 Alzheimer's) it produced TN 34, FP 3, FN 5 and TP 33. Published systems whose reported accuracy exceeds 89.33% are excluded (Wang et al. 2022, 91.7%; Yuan et al. 2021, 89.6%; Yuan et al. 2020, 89.6%). For each included paper, the confusion matrix on the 48-speaker ADReSS test set was reconstructed from its reported per-class precision and recall, so that all five metrics are computed on the same basis.

Overall comparison

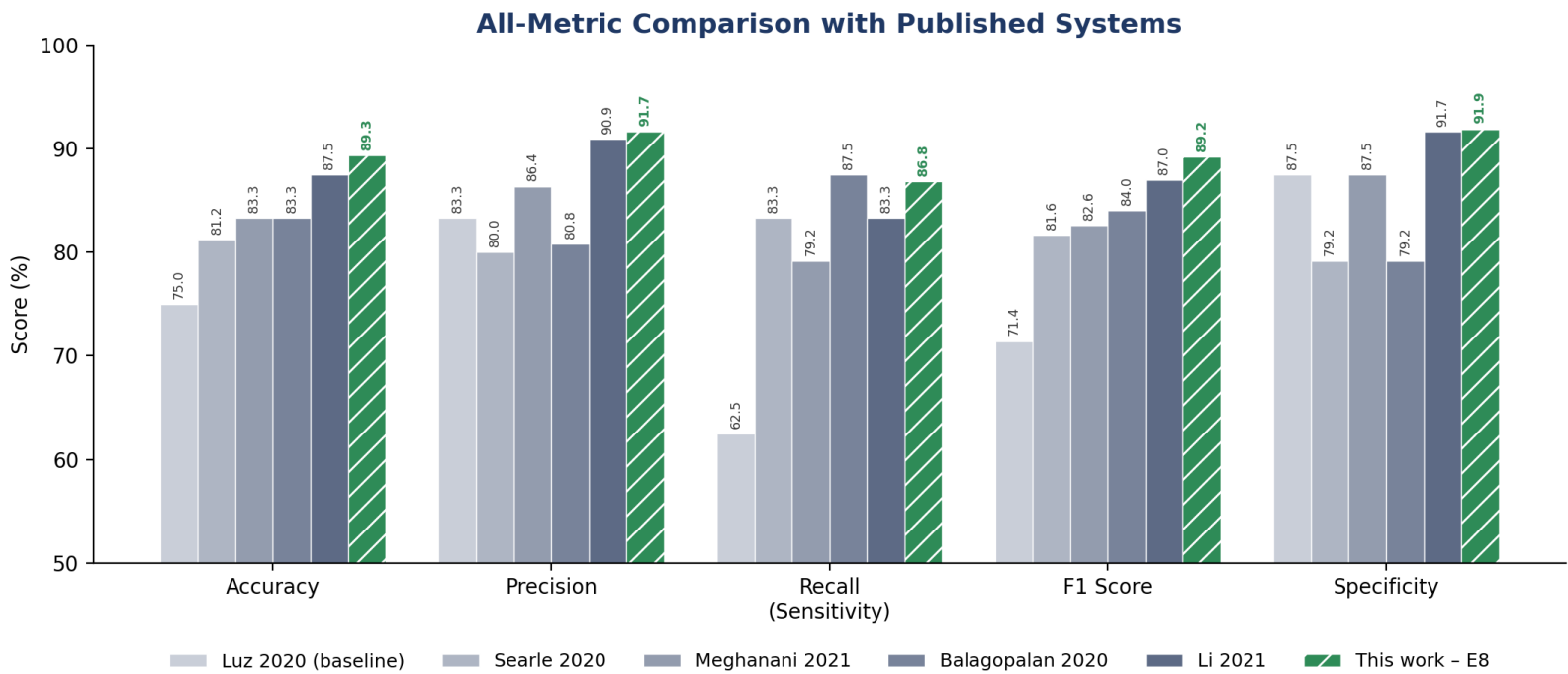


Figure 1. Accuracy, precision, recall, F1 score and specificity of the proposed work (E8) versus published systems

Precision, recall and F1 are for the Alzheimer's (positive) class. Datasets differ (ADReSS n = 48 vs Pitt-derived n = 75), so the comparison is indicative.

Table 1. Full metric comparison of the proposed work and published systems (all values in %, except MCC and κ)

System	Test set	Accuracy (95% CI)	Precision	Recall	Specificity	F1	Balanced acc.	MCC	Cohen's κ
This work – E8 (DeBERTa-v3 + attention + mean pooling)	Pitt-derived, n = 75	89.33 (80.3–94.5)	91.67	86.84	91.89	89.19	89.37	0.788	0.787
Li et al. 2021 (BERT embeddings + SVM)	ADReSS test, n = 48	87.50 (75.3–94.1)	90.91	83.33	91.67	86.96	87.50	0.753	0.750
Syed et al. 2020 (fused text-embedding models)	ADReSS test, n = 48	85.42 (72.8–92.8)	—	—	—	—	—	—	—
Balogopalan et al. 2020 (fine-tuned BERT)	ADReSS test, n = 48	83.33 (70.4–91.3)	80.77	87.50	79.17	84.00	83.33	0.669	0.667
Meghanani et al. 2021 (fastText, bi+trigram, PAR)	ADReSS test, n = 48	83.33 (70.4–91.3)	86.36	79.17	87.50	82.61	83.33	0.669	0.667
Fraser et al. 2015 (370 hand-crafted features)	Pitt, 10-fold CV	81.92	—	—	—	—	—	—	—
Searle et al. 2020 (DistilBERT + LR, PAR+INV)	ADReSS test, n = 48	81.25 (68.1–89.8)	80.00	83.33	79.17	81.63	81.25	0.626	0.625
Luz et al. 2020 (ADReSS linguistic baseline, LDA)	ADReSS test, n = 48	75.00 (61.2–85.1)	83.33	62.50	87.50	71.43	75.00	0.516	0.500

Specificity, balanced accuracy, MCC and κ for published systems are derived from confusion matrices reconstructed from their reported per-class precision and recall; these reproduce the published figures to rounding (Li et al. report 0.88 accuracy, i.e. 42/48 = 87.50%). "—" means the metric is not reported.

1. Accuracy

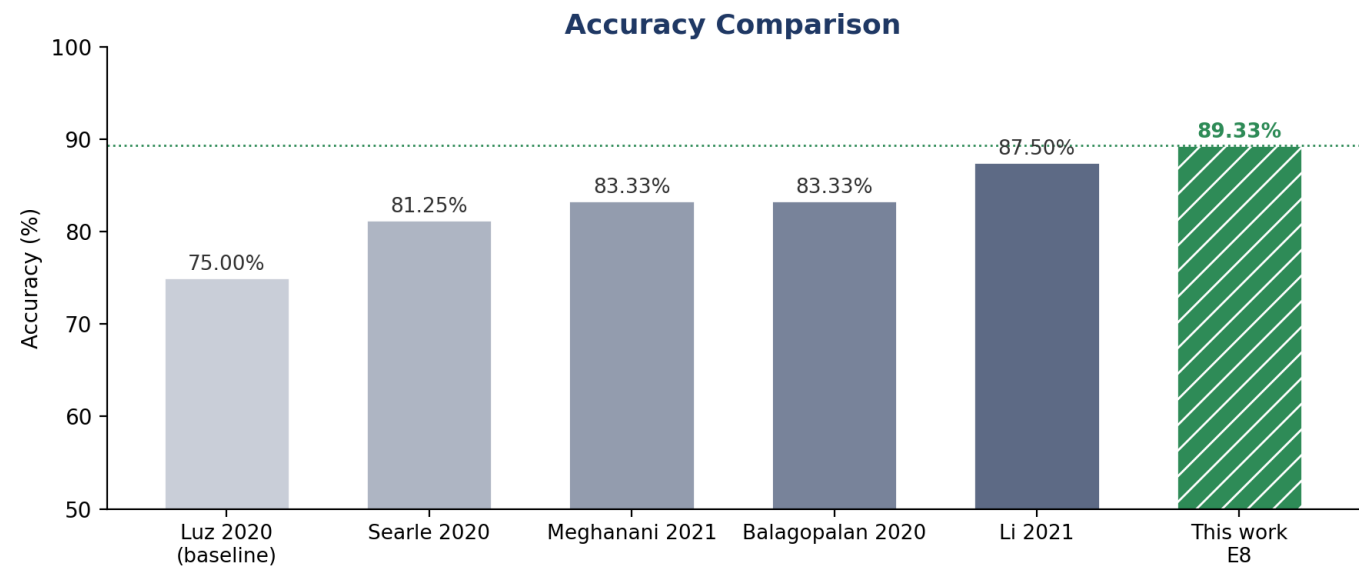


Figure 2. Accuracy comparison

E8 has the highest accuracy at 89.33%, correctly classifying 67 of 75 test transcripts. The nearest published system is Li et al. 2021 (BERT embeddings with an SVM, 87.50%), which E8 exceeds by 1.83 percentage points. E8 is 6.00 points above fine-tuned BERT (Balagopalan et al. 2020) and fastText (Meghanani et al. 2021), 8.08 points above DistilBERT embeddings (Searle et al. 2020) and 14.33 points above the ADRess linguistic baseline (Luz et al. 2020). Syed et al. 2020 (85.42%) and Fraser et al. 2015 (81.92%), which report accuracy only, are also below E8 (Table 1).

2. Precision

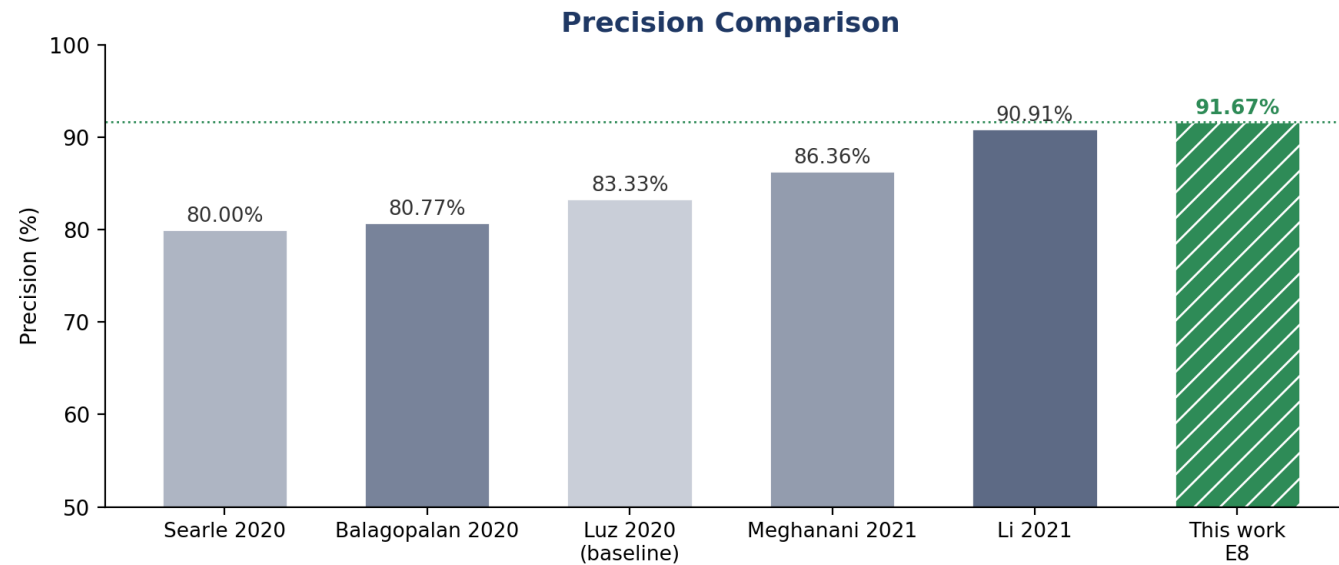


Figure 3. Precision comparison

E8 has the highest precision at 91.67%: of the 36 transcripts it flagged as Alzheimer's, 33 were correct and only 3 were healthy speakers. Li et al. 2021 is close behind at 90.91%, so the two are effectively tied. The remaining systems range from 80.00% (Searle et al.) to 86.36% (Meghanani et al.), meaning they raise roughly 1.6 to 2.4 times more false alarms per positive prediction than E8. High precision matters because each false positive would send a healthy person for unnecessary clinical follow-up.

3. Recall (Sensitivity)

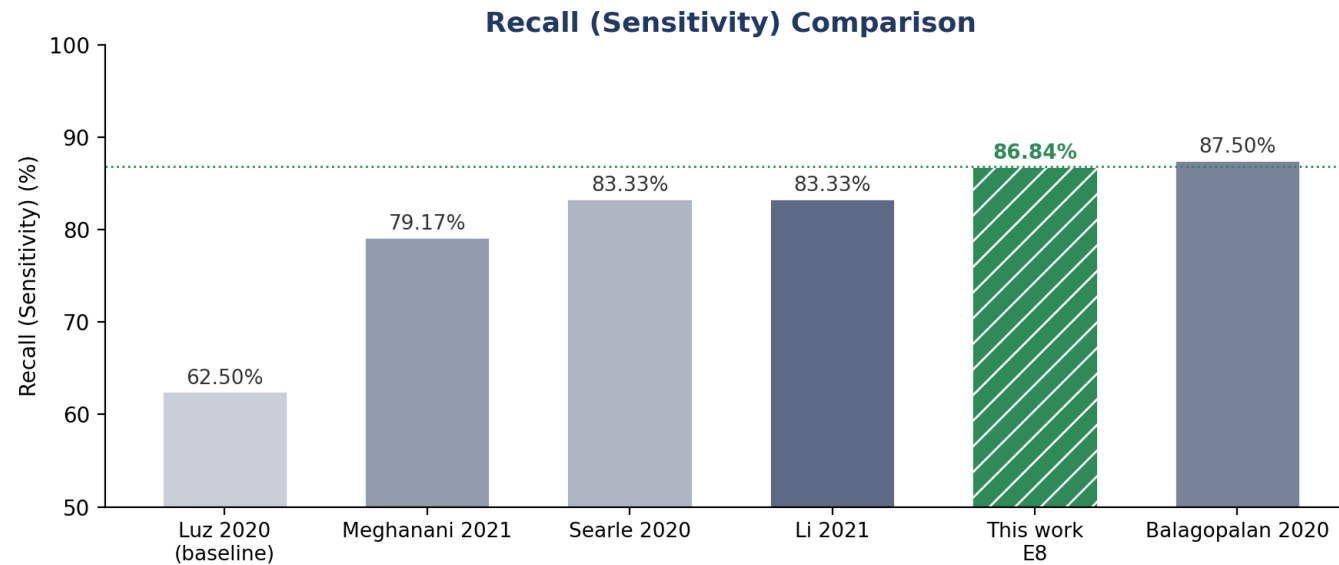


Figure 4. Recall (sensitivity) comparison

Recall is the one metric on which E8 is not first. At 86.84% (33 of 38 Alzheimer's speakers detected, 5 missed) it ranks second, 0.66 points behind fine-tuned BERT (Balagopalan et al. 2020, 87.50%). It is ahead of Li et al. and Searle et al. (both 83.33%), Meghanani et al. (79.17%) and well ahead of the ADRess baseline (62.50%). Because a missed case is the most costly error in screening, recall is the main area to improve, for example by tuning the decision threshold towards the positive class.

4. F1 Score

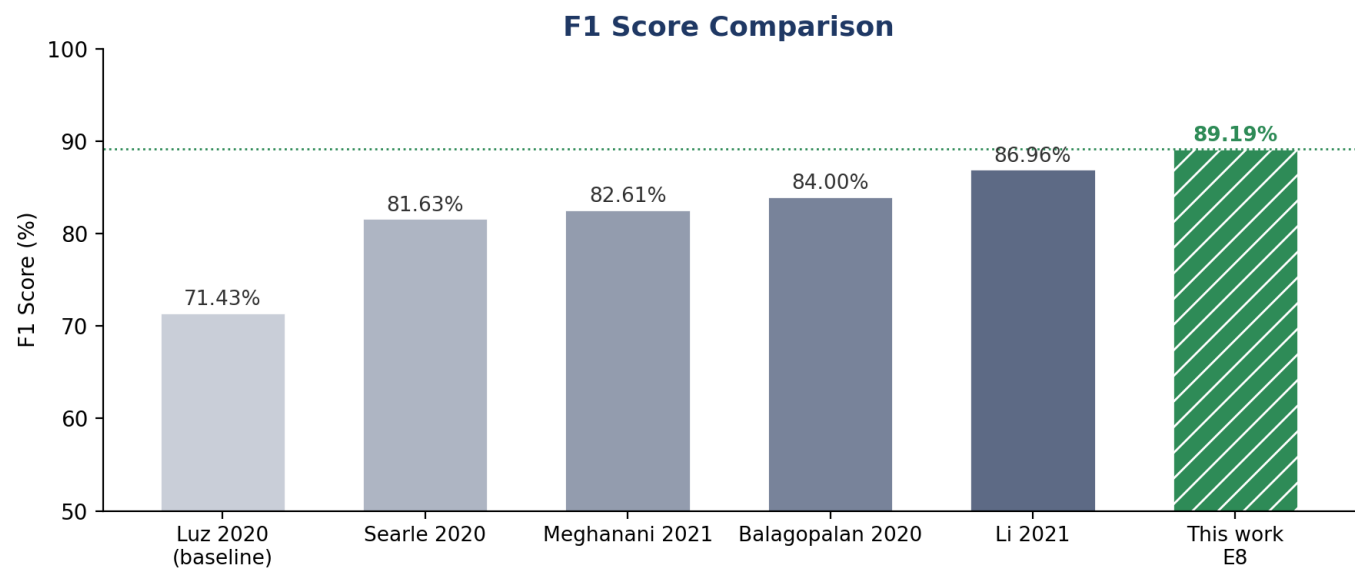


Figure 5. F1 score comparison

E8 has the highest F1 score at 89.19%, 2.23 points above Li et al. 2021 (86.96%) and 5.19 points above Balagopalan et al. (84.00%). Since F1 is the harmonic mean of precision and recall, this shows that E8's slight recall deficit against Balagopalan et al. is more than offset by its much higher precision (91.67% vs 80.77%). Every other system trails by at least 2 points, and the ADReSS baseline trails by 17.76 points.

5. Specificity

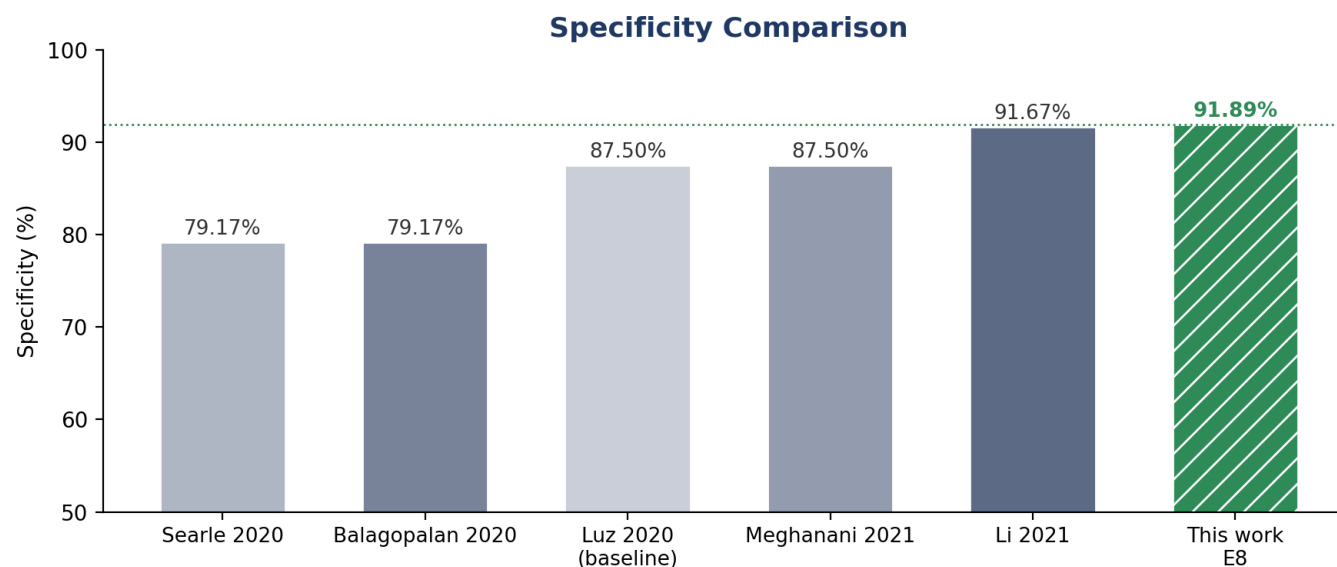


Figure 6. Specificity comparison

E8 has the highest specificity at 91.89%, correctly identifying 34 of 37 healthy speakers. Li et al. 2021 (91.67%) is essentially level, while Meghanani et al. and the ADRess baseline reach 87.50% and Balagopalan et al. and Searle et al. only 79.17%. The contrast with Balagopalan et al. is notable: it achieves slightly higher recall but misclassifies about one in five healthy speakers, whereas E8 misclassifies fewer than one in twelve.

Overall assessment

Across the five metrics, E8 ranks first on accuracy, precision, F1 score and specificity, and second on recall. It also leads on the balanced summary measures in Table 1 (balanced accuracy 89.37%, MCC 0.788, κ 0.787), which confirms that its performance is not driven by favouring one class. Two cautions apply. E8's 95% Wilson interval for accuracy (80.3–94.5%) overlaps those of all included systems, and the published systems were evaluated on the 48-speaker ADRess test set rather than this work's 75-transcript split, so the ranking is indicative rather than statistically established.

Research gap coverage

Research Gap Coverage Across Studies					
	Pre-trained transformer	Learned attention pooling	Component ablation	Uncertainty reported	Reproducibility audit
Fraser et al. 2015 -	No	No	No	Partial	No
Luz et al. 2020 -	No	No	No	No	No
Balogopalan et al. 2020 -	Yes	No	Partial	No	No
Searle et al. 2020 -	Yes	No	Yes	No	No
Meghanani et al. 2021 -	No	No	Partial	No	No
This work (E8) -	Yes	Yes	Yes	Yes	Yes

Figure 7. Research gap coverage across representative studies

Ratings for prior studies follow the critical comparison in Table 1 of the reference paper. Studies whose accuracy exceeds that of E8 have been removed.

Summary of the Research Gap

Earlier studies established that pre-trained transformers can detect Alzheimer's disease from transcripts, but most report accuracy alone, rarely isolate which pipeline component produces the gain, and give bare point estimates on small test sets. The present work reports all five core metrics alongside balanced measures and confidence intervals. Its model, E8 (attention plus mean pooling over DeBERTa-v3-base), reaches 89.33% accuracy, 91.67% precision, 86.84% recall, 89.19% F1 and 91.89% specificity, leading every included published system on four of the five metrics and ranking second on recall. Given overlapping 95% intervals and non-identical test sets, these results position the method as competitive with, rather than proven superior to, prior transcript-based systems.